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ORIGINAL ARTICLE

Healthcare professionals' perception of using a web-based reminiscence therapy to support person with dementia during the COVID-19 pandemic

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Abstract

Background Reminiscence therapy (RT) is the most common non-pharmacological treatment for dementia care. The therapy stimulates the senses to evoke memories having the potential to reduce Behavioral and Psychological Symptoms of Dementia (BPSD). Digital RT, such as web-based reminiscence therapy (WBRT), has the potential to support dementia care and reduce the caregiving burden.

Aims This study aimed to explore healthcare professionals (HCPs) perceptions of utilizing WBRT in institutionalized settings to support persons with dementia during the COVID-19 pandemic.

Methods A qualitative phenomenological descriptive study was adopted and guided by Graham's Knowledge to Action framework. Online training on the use of WBRT was conducted, followed by interviews with HCPs.

Results Four major themes were identified on the potential use of WBRT in dementia care, including usability and efficacy, impact on caregiving, capability of reducing BPSD, and. feasibility during COVID-19 social distancing.

Discussion This study recognized the potential use of WBRT to support the person with dementia during the pandemic in institutionalized settings.

Conclusion The knowledge generated from this study will guide the future application of WBRT to support dementia care in diverse healthcare settings.

Keywords Dementia · Reminiscence therapy · Digital · Web-based · COVID-19

Background

Behavioral and psychological symptoms of dementia (BPSD) are among the most serious complications of dementia. Approximately 80% of persons with dementia (PWD) will develop BPSD throughout their disease progression [1]. The clinical presentation of BPSD includes anxiety, depression, agitation (a state of anxiety), restlessness, aggression (hostile or violent behaviors), hallucination (unreal perceptions), delusions (false beliefs), apathy (lack of initiative), sleep disturbances, and wandering (moving aimlessly from place to place) [2].

BPSD management is highly challenging for healthcare professionals (HCPs) due to its complex etiopathogenesis of the signs and symptoms. Though no permanent solutions are available to manage BPSD and its progressive nature, several pharmacological and non-pharmacological treatments (NPT) are proven to be effective [3].

Reminiscence therapy (RT) is a form of NPT that support dementia care. It is a treatment that stimulates all the senses—sight, sound, touch, smell, and taste—to aid PWDs to remember events, people, and places from their past lives [4]. Such cognitive simulation approach can be facilitated using technology, such as digital photo albums or video

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chats with friends and family [5]. In contrast, other forms of cognitive stimulation using technology, such as brain-training games and puzzles, are designed to improve specific cognitive abilities, such as working memory or attention [6, 7]. These structured activities can be challenging, with the goal of strengthening cognitive skills over time. While both reminiscence therapy and other forms of cognitive stimulation using technology can be valuable tools for promoting cognitive health and well-being, their approaches and goals are distinct [5].

There is no formal categorization of RT; however, based on technology, this NPT can be divided into either traditional RT or digital RT. In traditional RT, HCPs interact with PWD through conversations or physical aids like photos or objects from the patient's past life. Digital RT follows a similar philosophy, but the mode of RT delivery uses some form of digital aspect, like a laptop, tablet, smartphone, Virtual Reality (VR) device, etc. Although traditional RT slows the progression of dementia, it has its limitations that need to be addressed. The effects of traditional RT are inconsistent and differ considerably across settings and modalities [4].

The way RT is delivered depends on the HCP's training and expertise. It also depends on the resources available at the institution. Also, HCPs must provide and repeat the dialog several times to remind the PWDs about their past with the traditional RT. Besides, items required for memory triggers must be collected and prepared before each session, which could be time-consuming and could be seen as a challenging caregiving responsibility [8]. These issues have aggravated since the COVID-19 pandemic.

It has also been identified from several studies that dementia care in institutionalized settings (particularly long-term care) has been disproportionately affected during the pandemic [9, 10]. The high-risk working environment led to a staffing shortage in such settings [11]. HCPs have to conduct a one-on-one sessions with the LTC residents for care delivery and social, recreational activities while taking extra precautions (such as frequent hand washing, and wearing Personal Protective Equipment). The social distancing protocol among residents of institutionalized settings also increased the caregiving burden of HCPs. Amidst such social distancing protocol; digital RT, such as web-based applications, has the potential to advance traditional RT.

Web-Based Reminiscence Therapy (WBRT) is a form of digital RT application where HCPs can use any device with a web browser to provide RT. The application allows HCPs to create separate accounts for each PWD in their care and upload media contents such as pictures, audio, and videos to access them during RT sessions. This feature of the application highlights the client-centered care options of RT as individualized content of PWD can be uploaded based on their personal life and preferences. The web-based application also has a generic photo album feature which can be used to deliver group RT among the PWD with similar interests. For instance, if a group of PWD like beaches, a generic picture of beaches can be displayed to them during the RT sessions.

Studies specific to utilization of WBRT are limited; however, digital RT provided through an android app has been reported to decrease the sense of loneliness, reduce the levels of anxiety and isolation, and improve the overall quality of life for PWD. Web-based applications have been found to increase interaction between PWD and caregivers [12]. Moreover, it has been suggested that digital RT may positively affect the cognitive behavior and emotional well-being of PWDs [13]. A recent systematic review has also advocated for the use of technology to facilitate cognitive improvement [14]. Since WBRT is a form of digital RT, it has the potential to be used as an alternative form of RT in institutionalized settings.

As discussed earlier, a limited number of studies have investigated technology-based RT in the treatment of dementia. To our knowledge, no studies specifically explore whether the web-based RT application can be a potential option for managing BPSD.

There is also a lack of emphasis on utilizing qualitative design to explore the perceptions of HCPs in utilizing digital RT to support dementia care. Further research needs to be conducted to examine the perceptions of HCPs about the potential benefits of WBRT application and to what extent this technology could potentially improve the HCP's capacity to manage BPSD, especially during the challenging time of COVID-19 pandemic.

The overarching research question of this study is:

"What are HCP's perceptions of using web-based RT applications to support the care of person with dementia with BPSD in the institutionalized setting during the COVID-19 pandemic and beyond?"

Methodology

Research design

This study adopted a qualitative descriptive phenomenological approach and grounded in the knowledge to action (KTA) framework by Graham et al. [15] to explore the lived experiences of HCPs about WBRT to support persons with dementia with BPSD [15]. KTA framework suggests that research findings can be transferred to practice by breaking them down into two concepts: knowledge creation and the action cycle. The KTA framework was utilized to describe the use of web-based application to advance reminiscence therapy experiences for PWD (Knowledge creation/KC) and to explore the potential impact of this technology on reducing BPSD (Action cycle/AC). This study obtained Ontario

Shores Centre for Mental Health Sciences and Ontario Tech University Research Ethics Approval (REB#19-011-B).

Study sample

We included HCPs that were employed in the Geriatric Dementia Unit (GDU) and Geriatric Transitional Unit (GTU) in Ontario Shores Centre for Mental Health Sciences, Whitby, Ontario, who were directly working with PWD.

Data collection

The HCPs were provided with an online education and training on WBRT using an online (Zoom) platform at a mutually agreed time. During the training, HCPs had the opportunity to explore the technology virtually. Due to time constraints, shortage of devices, and physical restrictions during the pandemic, HCPs could not try the application physically and independently. However, they were actively involved throughout the training session. After the initial demonstration of the application, the researcher requested the HCPs to ask questions and explore the WBRT interface. During this stage, the researcher navigated the application based on the HCPs command. This provided an opportunity for HCPs to explore the various features and functionality. After the virtual session, HCPs were interviewed about their lived experience with the WBRT and inquired about their perceptions of using this application in supporting dementia care. The HCPs verbally completed a demographic survey prior to the interview (Appendix-I).The estimated duration of the interview was approximately 30–45 min and was guided by an interview guide (Appendix-II) which was comprised of semi-structured open-ended questions.

Description of the technology

This WBRT application was created by a team of researchers from Ontario Tech University in collaboration with the community partners from Ontario Shores Centre for Mental Health Sciences in Whitby, Ontario [16]. HCPs of Ontario Shores were involved in the co-designing of the WBRT. During the early stages of the development, a research team from Ontario Tech University sought feedback from HCPs to explore user needs and preferences related to the features and functionality. Their feedback and suggestions were later incorporated into the development of the WBRT.

The WBRT can be accessed by the HCPs through a link and log-in page using an individual username and password. The HCP can then have access to the photo album containing generic/personalized pictures or videos suitable to support reminiscence therapy for PWD. There were wide range of pictures and videos in an album format to choose from. A sample webpage is displayed below. HCPs were able to select one of the photo albums and deliver WBRT as a trial. A generic picture has been shown below as an example (Fig. 1).

The WBRT can be exhibited via monitors (TV, desktop, laptop, tablet, iPad). HCPs can upload password-protected, generalized/ personalized content about their patients according to their preferences and descriptions of patient profiles. Additionally, HCPs can download the archived contents by logging into their individual accounts and album can be transferred via email or save as a PDF file. These downloaded contents can then be shared among direct care providers when patients are discharged/transferred to other setting.

Data analysis

The data analysis process was grounded within Braun and Clarke's [17] six steps thematic analytical model using an excel codebook. It consists of (1) Familiarization of data (2) Initial coding (3) Generating themes (4) Validity and reliability of themes (5) Defining and naming themes (6) Interpretation and reporting.

Results

A total of ten HCPs were recruited, which consisted of occupational therapists ($n=2$), behavioral therapists ($n=2$), recreational therapists ($n=2$), social workers ($n=2$), and nurses ($n=2$) (Appendix III).

Qualitative thematic findings

A total of four major themes were identified related to the application and usability of the WBRT application. The qualitative findings are summarized in the thematic tree (Fig. 2). Relevant quotes from the interviews were summarized in the Appendix IV.

(1) **Usability and efficacy of WBRT**

(a) ***User-friendly and inclusiveness*** HCPs found the user interface of WBRT was simple and could be easily navigated by people who knew how to use the internet. The HCPs expressed that the WBRT's user interface allowed both HCPs and family members to participate in the reminiscence therapy experience. Family members can be more engaged with caregiving by uploading customized content to facilitate RT for their loved ones. In the absence of a family member the HCPs can create the account in the WBRT and upload the reminiscence contents with generic pictures. (Refer to Appendix-2 Q# 1)

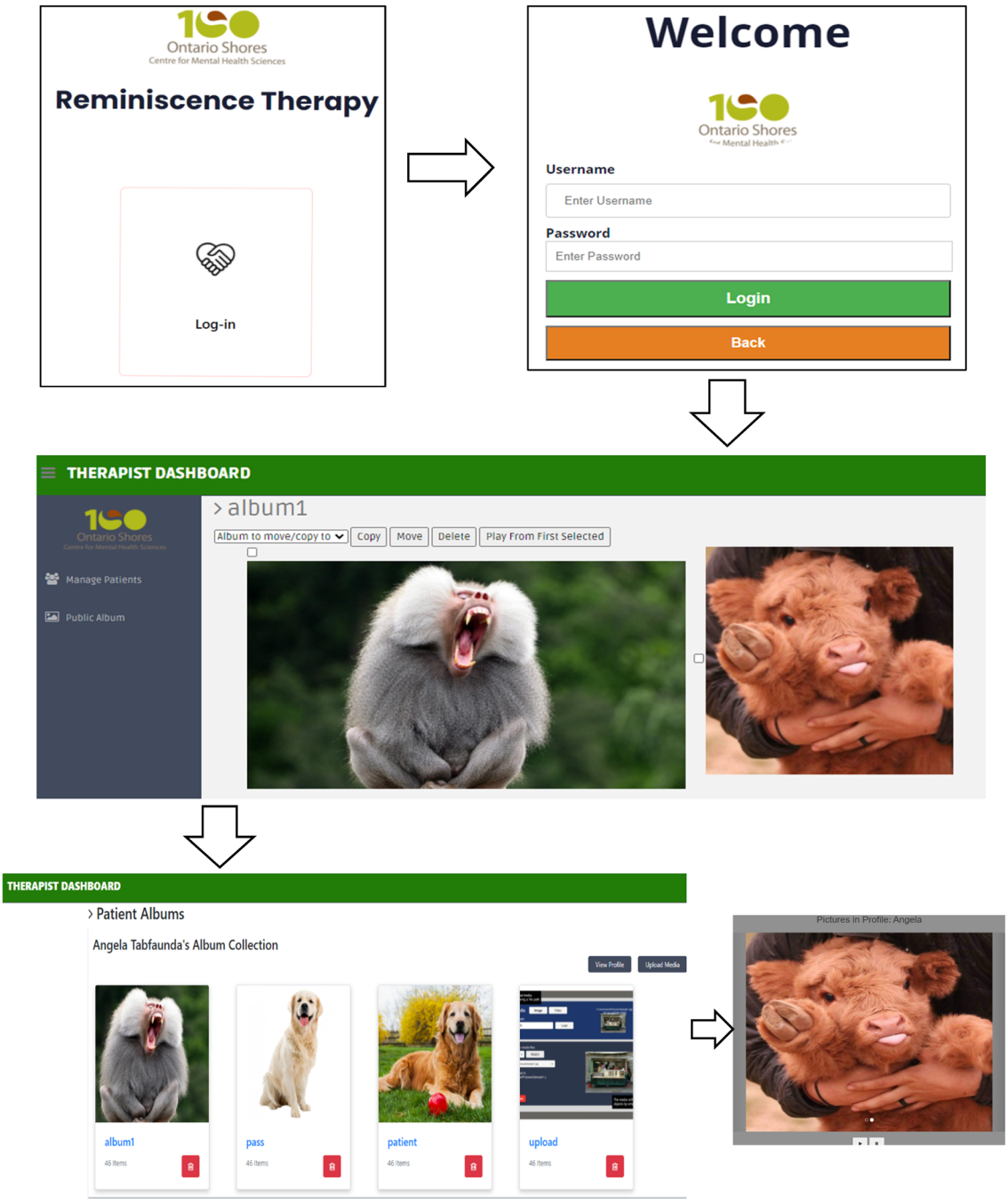

Fig. 1 Web-based reminiscence therapy application

(b) ***Client-centered and individualized approach to therapy*** The HCPs believed that PWD needs individualized care and providing them with memory triggering content is of immense importance to promote their sense of well-being. HCPs agreed that WBRT provided a platform to upload customized content (photo, video, music, etc.) tailored to each individual's needs. This approach added value to their care and increases engagement, and PWD may connect more with the content than traditional methods. (Refer to Appendix-2 Q# 2, 3)

(c) ***Digitalization of RT*** HCPs agreed that digitalization provided an opportunity to store more information and organize the materials needed for RT. They believed that WBRT would potentially reduce their need for paper copies and help them keep all client information in an accessible and secure place. They also agree that RT needs digitalization to make the process more

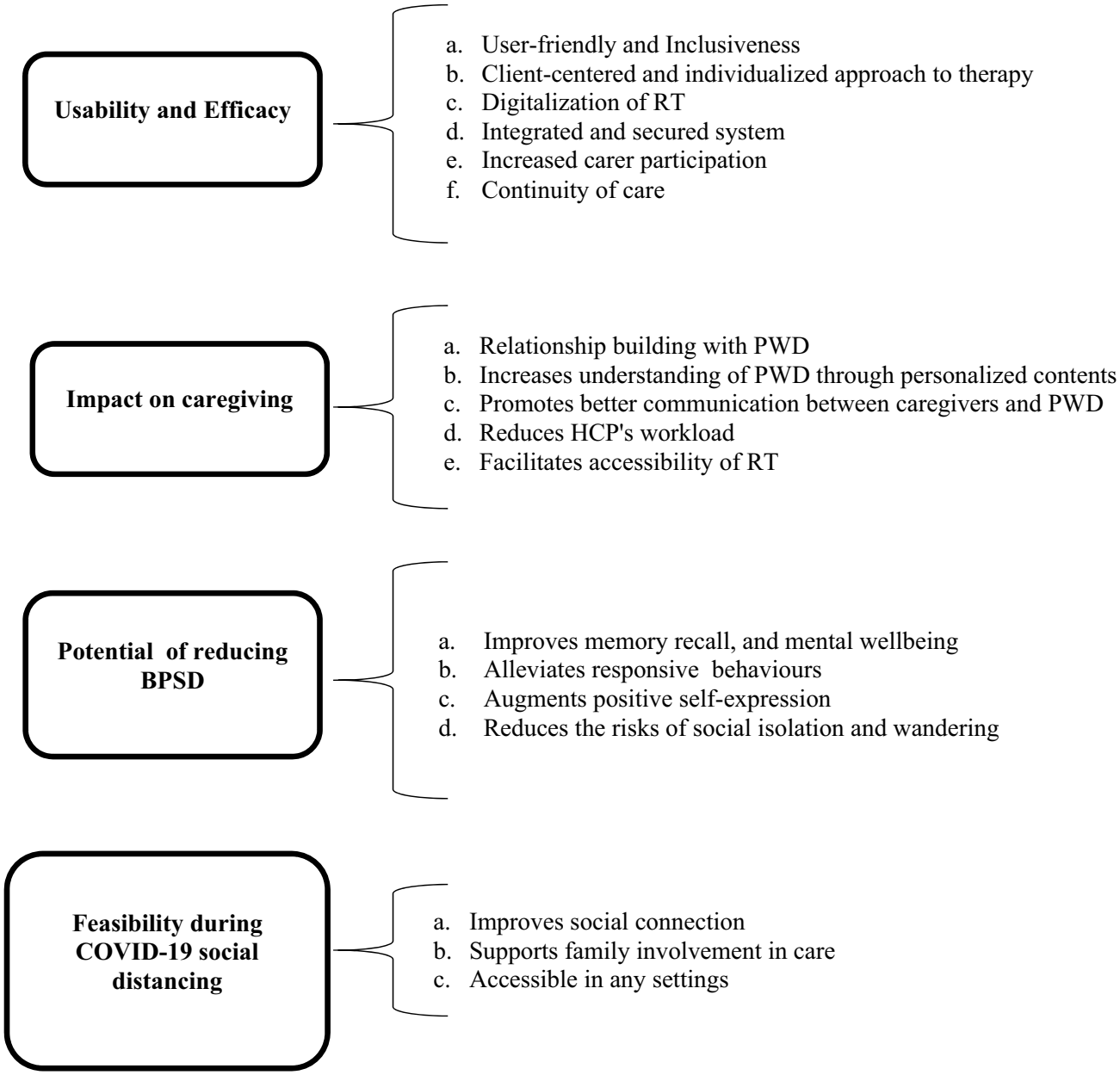

Fig. 2 Thematic tree

efficient, streamlined and accessible to them. (Refer to Appendix-2 Q# 4)

(d) ***Integrated and secured system*** The WBRT includes pictures, music, and videos for each patient profile under one platform. HCPs can upload and access an individual patient's content without the need to navigate multiple platforms and places. This integrated feature reduces the need for HCPs to arrange RT contents before every session. Alternatively a printed copy might get lost in an institutionalized setting. Thus, using the integrated digital platform is more secure and can maintain patient privacy and confidentiality. (Refer to Appendix-2 Q# 5, 6)

(e) ***Increased carer participation*** Once a patient account and content have been set in the WBRT, anyone with access to the technology, such as family members, volunteers, or HCPs, can view and deliver the RT. Traditional RT is challenging and time-consuming for both family members and HCPs. For instance, to provide the traditional RT using Snoezelen (multi-sensory therapy), one needs to be formally trained and certified before providing the therapy. (Refer to Appendix-2 Q# 7)

(f) ***Continuity of care*** Family members or loved ones can upload artifacts remotely and update the album according to the needs of their loved ones. HCPs no longer need to wait for the hard copies of pictures or videos to be delivered for the therapy; this will promote the continuity of the caregiving process in a timely manner. In addition, the patient album could be transferred via secured email or downloaded and saved as a PDF file by authorized personnel. This feature would aid in sharing RT contents among facilities and assist caregivers in continuing the therapy with their loved ones at home or in other healthcare settings. (Refer to Appendix-2 Q# 8)

(2) **Impact on caregiving**

(a) ***Relationship building with PWD*** The HCPs indicated that the WBRT could potentially increase the connection between HCPs and PWD. Due to the personalized contents of the WBRT and its client-centered features, this could help HCPs know their patients on a more personal level and help to strengthen the therapeutic rapport. (Refer to Appendix-2 Q# 9)

(b) ***Increases understanding of PWD through personalized contents*** WBRT could aid HCPs in increasing their understanding of their patients, such as personal background and history. Viewing individual albums with images, music, and videos will enhance HCP's understanding of the patient prior to the implementation of RT. HCPs would be able to better connect with PWD to facilitate dementia care. During the interview, all the HCPs preferred a personalized album over the public album, and they indicated that personalized content would be more relevant and meaningful to PWD than generic content. (Refer to Appendix-2 Q# 10)

(c) ***Promotes better communication between caregivers and PWD*** Communicating with PWD can be challenging for HCPs during the disease progression. Due to the reduced receptive and expressive communication skills, PWD can experience difficulty in interacting with other people. The HCPs indicated that the WBRT could generate meaningful conversation and facilitate effective communication with PWD. (Refer to Appendix-2 Q# 11)

(d) ***Reduces HCP's workload*** PWD needs extra time and attention for care, which would likely increase HCP's workload. The WBRT could reduce HCPs' workload as they can retrieve all the personalized contents of PWD in a single platform. HCPs highlighted the simplicity and efficiency of the WBRT and how it could ease their caregiving responsibility by knowing their patients' needs and preferences. (Refer to Appendix-2 Q# 12)

(e) ***Facilitates accessibility of RT*** In a dementia-care institutional setting, each PWD could be managed by multiple HCPs throughout their stay. The WBRT could act as a repository of each patient, and staff can access the content whenever required. The HCPs do not have to collect the reminiscence contents each time prior to the therapy. Any staff could log into the WBRT using any device, which makes the delivery of RT more accessible to PWD at anytime and in any location. (Refer to Appendix-2 Q# 13)

(3) **Potential of reducing BPSD**

(a) ***Improves memory recall and mental wellbeing*** The HCPs indicated that WBRT application could be used to help recall good memories of PWD. When being away from home and in a unfamiliar environment, PWD might get anxious and become disconnected. By viewing personal images and videos, PWD could feel more connected with their family members, and thereby improving their mental wellbeing. (Refer to Appendix-2 Q# 14)

(b) ***Alleviates responsive behaviors*** During the interview, it was identified that WBRT may positively impact BPSD. The WBRT can potentially alleviate responsive behaviors such as anxiety, depression, restlessness, and exit-seeking behaviors among PWD. HCPs discussed the potential of WBRT in reducing irritability and physical and verbal agitation. (Refer to Appendix-2 Q# 15)

(c) ***Augments positive self-expression*** PWD often deals with restlessness and distress, impacting their confidence and quality of life. HCPs indicated that the WBRT could potentially instill a positive mindset and enhance self-expression by connecting PWD with their loved ones through the review of their life histories during reminiscence experiences. (Refer to Appendix-2 Q# 16)

(d) ***Reduces the risks of social isolation and wandering*** Due to the decrease in social interaction and communication with others, PWD often feels lonely and isolated. HCPs stated that the WBRT could potentially reduce loneliness and risks of wandering by connecting their past lives to the present through the narratives of reminiscence journey. (Refer to Appendix-2 Q# 17)

(4) Feasibility during COVID 19 social distancing

(a) ***Improves social connection*** COVID-19 social distancing has impacted the life of PWD and their informal caregivers, such as family members or loved ones, by reducing social connection. Due to visitor restrictions, family members were not allowed to visit their loved ones in institutionalized settings during certain stages of the pandemic. The WBRT has the functionality to virtually connect families from a distance, thereby enhancing their social relations and potentially reducing their sense of loneliness. (Refer to Appendix-2 Q# 18)

(b) ***Supports family involvement in care*** Due to the social distancing protocol, it was challenging for family mem-

bers to deliver reminiscence contents in-person. HCPs indicated that through WBRT, family members could upload personalized content remotely at their convenience allowing them to positively contribute and get involved in the care and treatment of their loved ones. (Refer to Appendix-2 Q# 19)

(c) ***Accessible in any settings*** The WBRT can store the digital individualized contents to support reminiscence therapy (i.e., audio-visual prompts) and it be accessed remotely from any setting using the internet. (Refer to Appendix-2 Q# 20)

Discussion

The primary purpose of this study was to explore HCPs' perception of the usefulness of WBRT to assist reminiscence experiences in institutionalized settings. Studies have found that innovative technologies can be used to provide RT [5]. Similarly, the digital technologies offer availability and multi-user access to the patient file in real-time, which leads to simplification and acceleration of healthcare processes. It also improves care coordination and optimizes health care management costs [18]. Once the WBRT account has been developed for a particular patient, staff, volunteers, and family members could view the patient profile, background and history to guide the RT. Such functionality is beneficial for RT as it boosts consistency and continuity of dementia care [19].

Our current study found that HCPs may connect better with PWD through WBRT by developing a therapeutic rapport with them. The personalized album options of WBRT could aid HCPs in knowing their patients' personal lives and preferences in a timely manner, thereby facilitating client-centered care. Client-centered care is a significant principle for dementia care as it prioritizes PWDs' well-being, promotes the strengths of PWD's, and honors their choices and values [20]. Additionally, client-centered care focuses on the unmet needs of PWD and can create positive patient outcomes, including better quality of life, reduced agitation and challenging behaviors [21].

An earlier study by Samuelsson and Ekström (2019) examined the application of personalized versus generalized content using web-based applications in a residential home facility. They video recorded and analyzed the interaction between PWD ($n = 6$) and nurses ($n = 3$) and conducted a semi-structured interview with PWD to evaluate their web-based application experience [22]. Our study expanded the scope of previous research by including HCPs from multiple disciplines, including recreational therapists, occupational therapists, behavioral therapist, social workers and nurses and found similar results which underscore the practical application of digital RT.

Digital technology can support PWD to combat their disease progression. It also has a significant influence on their mental health and well-being by managing BPSD symptoms. A case study by Rehm et al. (2016) has identified that web-based applications would be feasible among PWD in residential aged care facilities. It has the potential to improve cognitive behavior and reduce depression and anxiety [23]. Our current study recognized that WBRT could potentially make a positive impact on improving mental health and positivity among PWD through the multi-sensory stimulation offered by reminiscence experiences in institutionalized healthcare settings.

WBRT could store all the media in a single platform and be navigated through multiple devices such as laptops, desktops, and tablets using the internet, which promotes the accessibility of the intervention to be used in any healthcare setting. The importance of accessibility of technology in healthcare services is critical, as it provides consistency and continuity of care and better patient management. It can also decrease human errors, increase clinical outcomes, facilitate care coordination, and improve practice efficiencies [24].

Limitations of the study

The current study evaluated the potential use of WBRT from a small sample of HCPs ($n = 10$), which can be considered a study limitation. Although the data have been collected from a variety of HCPs such as nurses, occupational, recreational, and behavioral therapists, and social workers, it did not include the perceptions from the physicians and the family members, who may be able to offer additional perceptions regarding the utilization of WBRT to support dementia care during the COVID-19 pandemic**.** In addition, the study was conducted in only one healthcare setting; therefore, the transferability of study findings may be limited. Future research expanding the sample size to include HCPs from diverse healthcare settings to better understand the various HCP's perceptions regarding the applications of WBRT in dementia care may be of benefit.

The researcher conducted the interview immediately after the technology training; thus, HCPs didn't have the opportunity to explore and navigate the application by themselves. Future research should focus on assessing user experiences once HCPs adopt WBRT as part of their routine dementia care.

Future implications

Our findings explore the potential role of utilizing a web-based application to support RT delivery to PWD in an institutionalized setting. The WBRT is simple to use and can be accessed through any types of digital device, which could potentially be used by the HCPs in any healthcare setting. The study findings highlighted the need for improving the accessibility of WBRT by implementing appropriate educational training among HCPs to promote increased utilization of the technology. Overall, findings from this study provide support of the value of WBRT, as well as some opportunities for future developments to optimize the delivery of RT and dementia care.

Conclusions

Our study results suggest that the WBRT application can be used as an effective intervention in dementia care. HCPs perceived that WBRT has the potential to reduce caregiving burden and improve the management of BPSD. They also highlighted that WBRT could play an important role in better supporting dementia care during and post pandemics.

This study provides a starting point for future large-scale intervention research to understand the impacts of WBRT in patient outcomes. It also provides guidance to policy makers on the critical factors that needs to be taken into consideration prior to implementing digital RT or similar technologies. Future research should examine the benefits of integrating both web-based application and immersive virtual reality technology to enhance the reminiscence experiences for PWD to support dementia care in diverse healthcare settings. Our study findings provide implications to stakeholders and policy makers on transforming the traditional RT therapy through digital technology to support technology-enabled dementia care during the current pandemic and beyond.

Appendix

Appendix I: Socio-demographic data form for study participant

Participant number	Date
.	.

Please tell us the following information about yourself

(1) Please check one of the following categories that apply to you

I am a Registered Nurse ☐ Registered Practical Nurse ☐ Recreation Therapist ☐ Personal Support Worker ☐

Other ☐

(2) What is your age range?

18–29 ☐, 30–39 ☐, 40–49 ☐, 50–59 ☐, 60–69 ☐, 70+ ☐

(3) What is your gender?Male ☐ Female ☐ Other ☐

Prefer not to answer ☐

(4) What is the highest level of education completed?

(a) Some grade school/high school ☐
(b) High School Diploma/ GED ☐
(c) College Diploma ☐
(d) University (i.e., BscN) ☐
(e) Graduate program (i.e., Master's; PhD) ☐
(f) Prefer not to answer ☐

(5) What is your employment status?

(a) Full-time ☐
(b) Part-time ☐
(c) Casual ☐
(d) Retired ☐

(6) For how many years are you working as a healthcare professional?

(7) How many years of working experience in institutionalized care settings do you have? _____
(8) Do you have any formal training in working with patients with dementia? If yes, please define: (i.e., certification in gerontology; cognitive rehabilitation etc.) _____
(9) How many years of working experience with persons with dementia do you have? _____
(10) How many years of experience do you have with traditional reminiscence therapy? ——-

Appendix II: Interview guide

Interview prompt questions

(1) What prompted you to agree to this interview?
(2) Tell me about your experience of reminiscence therapy in managing BPSD? (Types of RT you have been using? Advantages and disadvantages of it.)
(3) Did you have experienced using technologies for reminiscence therapy before? If yes, what is your preferred method of delivering reminiscence therapy among PWD (traditional/digital)?
(4) What is your overall impression of using web-based reminiscence therapy application?
(5) How far do you think using the web-based application may affect your relationship with PWD? Why?
(6) What impacts do you anticipate with the use of web-based reminiscence therapy on your caregiving responsibilities?
(7) Do you think using this digital technology during COVID-19 social distancing is feasible? Why or why not?
(8) What are your thoughts on substituting the web-based application with traditional RT at institutionalized settings? Why?
(9) Do you think PWD could use this technology for self-management? Why or why not?
(10) Did you experience any difficulties in using web-based applications? What were the challenges?
(11) What are your perceptions of potential facilitators in implementing web-based applications in dementia care?
(12) What are your perceptions of potential barriers in implementing web-based applications in dementia care?
(13) What is your opinion on using personalized content (picture/ video) as RT for PWD? Do you think this would be helpful? Why or why not?
(14) Will you recommend others to use this web-based technology as reminiscence therapy? Why or why not?
(15) Is there anything else you would like to say that I have not asked about?

Appendix III: Descriptive summary of demographic data

				Experience working (in years)			
Participant	Age range	Gender	Education	As HCP	In institutionalized settings	With PWD	With traditional RT
OT1	18–29	Male	Graduate program	3	3	3	No
OT2	18–29	Female	Graduate program	3	2.5	2	No
BT1	30–39	Female	Graduate program	5	2	1	No
BT2	40–49	Female	University (BscN)	22	22	15	Yes
RT1	30–39	Female	University (BscN)	7	6	2	Yes
RT2	40–49	Female	College diploma	25	25	20	Yes
SW1	30–39	Female	Graduate program	8	3	4	No
SW2	50–59	Female	Graduate program	20	20	7	Yes
N1	40–49	Male	University (BscN)	12	12	12	No
N2	30–39	Female	University (BscN)	14	14	14	No

Appendix IV: Quotes from the thematic analysis

Usability and efficacy of WBRT

User-friendly and inclusiveness

Q#1: "I think it's incredibly user-friendly. I think it's super well thought out. I think it certainly meets the needs of our patients here at Ontario shores, and I really like the fact that not only can an account be made through a staff, but a caregiver can also participate in that form of engagement, and then if patients don't have caregivers or families …. at least staff still have the opportunity to be able to make an account for the patient and include things that are maybe more

public based pictures like you were showing..,.... So, I like that they're not leaving out everybody. It's kind of inclusive to everybody." [OT2]

Client-centered and individualized approach to therapy

Q#2: " I think you can certainly be more in touch with families if the families are technically savvy and have the ability to draw on personal experiences, memories, preferences, and individuality. I think it is important, and it's more meaningful for them when they go through something like that, and it speaks to them. Like I said, diversion and distraction are critical when it comes to managing BPSD, and if you can catch the person's attention and maintain it for any length of time, then that's a success. So, and I do think that use of this kind of technology would certainly work towards that goal." [BT2]
Q#3: "What I see and what I've seen so far, it (WBRT) looks really good. So, I would recommend the use of it because it's just so much easier for the caregiver to have the ability to upload photos. We can also upload images, videos, whatever they provide us. It's very client-centered, so it's definitely a great tool to use." [OT1]

Digitalization of RT Q#4: "I think that (WBRT) definitely an advancement, because currently we're going more digital on how I'm performing my reminiscence therapy, and our patients are able to connect on the larger screen with that, so it will support, especially when you look at a lot of our aging population now is at a younger age where they connect with technology anyway, so it would be accessible for them" [RT1]

Integrated and secured system Q#5: "I'm all for it, I think it's a better idea because it's much easier to be able to bring somebody on an iPad or a computer that everybody shares. That's kept in the same spot but has everybody's information in separate folders. Rather than having multiple pieces of paper, or multiple books lying around, so I think it will be much more organized and you know the videos, the pictures, everything will all be there together. So it's not coming from different places." [RT2]
Q#6: "You know, as being in a hospital setting, it's always preferred not to have things in print laying around with patient's personal information. So I think if it's all (information) just embedded within technology...it makes it a little bit more secure." [BT1]

Increased carer participation Q#7: "I think it (WBRT) can save a lot of time …, the patient will still require to have one-on-one person to be with them, and go through the information, but I think… somebody like volunteer for example, or a staff member or family member would be able to conduct and support being able to view the reminiscence therapy, whereas the more traditional methods like Snoezelen, for example, it does require people to be formally trained and certified before they're able to operate certain functions of this Snoezelen so that in some ways we'll be able to open up the opportunity for a lot more participation from various parties."[SW1]

Continuity of care Q#8: "One of the barriers is getting to know the person before their dementia. Because before, I relied so heavily on seeing the family weekly, where now there's that disconnect.… On more of a personal level and with the content being able to be uploaded regularly, I can learn more what kind of interventions or images work best for them and what they connect to most and maybe what ones see a disconnect. I'm on a council where we're looking forward at what dementia care within our facility is going to look like, and one of the big things from a recreational therapy perspective is that I've realized there's a deficit in being able to pass on information and toolkits about a patient to their next facility. So when they're returning back to the long-term care home, if I could give the caretaker that information to send along, they could continue to utilize that intervention for the patient."[RT1]

Impact on caregiving

Relationship building with PWD

Q#9: "I think it (WBRT) will benefit our relationships. I think it will not only provide engagement for our patients, but it will also allow us as staff to get to know our patients on a deeper level, which can help with overall empathy and understanding of where the patient came from or what their background is about, so I think it'll strengthen therapeutic rapport."[OT2]

Increases understanding of PWD through personalized contents

Q#10: "I think that a lot of times people with severe dementia, we are not able to get to know them in ways where they (PWD) can express themselves. So I think it (WBRT) allows a practitioner to be able to learn more

about the patient through the media that they're viewing."- [SW1]

Promotes better communication between caregivers and PWD

Q#11: "Yes, I do think it (Personalized WBRT) would be helpful. I think both receptive and expressive communication is often impaired in dementia and any other use that you can use to augment what you're trying to say. So whether it's words and visual stimuli like pictures is going to help you get your meaning across."[BT2]

Reduces HCP's workload

Q#12: "I think it's (WBRT) great as you're learning more about the patient because they're only technically here for 59 days. It does go by quickly. I know it does seem like a long time, but it goes by super quickly, so the quicker you know the patient their likes, hobbies, interests, and dislikes. You're building that therapeutic rapport to make treatment a lot easier hopefully, I really like that you get to learn more about the patient, and I think that it really just speeds up the work.. and it makes your job a little bit easier too."[OT1]

Facilitates accessibility of RT

Q#13: "I think this (WBRT) is helpful because everybody would have access to it at once. I could have it on a computer, and then if an additional staff had to view it, they wouldn't have to look for it. It would just be on the computer as well, and you wouldn't have to worry about losing it either."[N1].

Potential of reducing BPSD:

Improves memory recall and mental well-being

Q#14: "It (WBRT) will help… maybe recalling fond memories. I think that it can…, improve quality of life." [OT1]

Alleviates responsive behaviors

Q#15: "Like it (WBRT) will help reduce physically responsive behaviors, verbal aggression.. I think it can also help with more, so just anxiety and anxiousness and then kind of reduce physical, verbal response behavior such as being anxious, restlessness and having psychomotor agitation."[OT1]

Augments positive self-expression

Q#16: "So some of the symptoms that I notice most is exit seeking behavior, and I could see having the reminiscence therapy as a way to alleviate that, so if they're able to connect with something familiar, you would see a decrease in that behavior. And with that also comes distress, right? So a decrease in distress, such as wanting to be at home and being in an unfamiliar environment. If they were able to look at images from their house or images of their loved ones. We would see more of positive self-expression coming from patients."[RT1]

Reduces the risks of social isolation and wandering

Q#17: "It (WBRT) could potentially help with feelings of loneliness and the wandering as well. So if they are wandering, they might be looking for their mom or looking for their sister or brother or their son. Yeah, I think it could help redirect them. It could help also reassure them."[BT1]

Feasibility during COVID 19 social distancing

Improves social connection

Q#18: "Absolutely, I think it's feasible because especially when we're looking at it being a web-based application where families from a distance can apply the personalized images or from their cellular devices, it allows for them to connect with their loved ones even from a distance. And I think with the added use of technology on our unit as well, if there was a loved one wanting to view how that reminiscence therapy is working for their loved one and see their reactions, we also would be able to use our other web-based connections with the family to connect them with that experience."[BT1]

Supports family involvement in care

Q#19: "I think it is super easy to use, and obviously with COVID-19, you're trying to social distance from one another, and you don't need to have family members directly coming in to drop off physical pictures, videos, whatever. It may be, like I said, they can upload it from home wherever they are and then you can get that and provide it to the client. So definitely very feasible for family members". [OT1]

Accessible in any settings

> Q#20: "Yeah, I think that it will be more helpful during COVID because, with reminiscence therapy, caregivers can potentially be putting material together from the comfort of their own home and then being able to use it anywhere else. Once the material is put together, the staff members, either in a long-term care setting or on our current unit, would be able to provide the support in a socially distanced manner."- [SW1]

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Informed consent All participants provided informed written consent prior to their participation.

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